

Basics for Biostatistics

Zeytu Gashaw Asfaw (PhD)
Department of Biostatistics and Epidemiology
School of Public Health, Addis Ababa University
Addis Ababa, Ethiopia

June 29, 2023

Table of contents

- 1 Introduction
- 2 Probability distributions (Normal, Binomial, Poisson)
- 3 Types of Sampling Methods
- 4 Sample Size Calculation
- 5 Sampling Distribution
- 6 References

What is Statistics?

- Statistics is a science deals with collect, organize, analyze, and draw meaningful inferences from data, which lead to good decisions.
- Statistics: is the art and science of making decisions in the face of uncertainty
- The field of statistics provides some of the most fundamental tools and techniques of the scientific method;
 - forming hypotheses,
 - designing experiments and observational studies,
 - gathering data,
 - summarizing data,
 - drawing inferences from data (e.g., testing hypotheses)

Stages in Statistical Investigation

- Data Collection
- Organization and Presentation of data
- Data Analysis
- Interpretation of the results

Limitation of Statistics

- It does not study qualitative characteristics directly
- It doesn't deal with a single individual but deals with aggregate of facts.
- Statistical findings are approximate
- It is sensitive to misuse.

Classification of Statistics

- Descriptive Statistics

- It helps to describe a given set of data without going beyond that data
- It consists of collection, organization, summarization, and analysis of data

- Inferential Statistics

- It helps to make inference/conclusion about a population based on the selected sample
- It consists of predict and forecast values of population parameters, test hypothesis about values of population parameters and make decisions

What is Probability?

- Probability is the branch of mathematics concerning numerical descriptions of how likely an event is to occur, or how likely it is that a proposition is true.
- Probability theory, a branch of mathematics concerned with the analysis of random phenomena.
- The outcome of a random event cannot be determined before it occurs, but it may be any one of several possible outcomes.
- The actual outcome is considered to be determined by chance.
- The probability of an event is a number between 0 and 1, where, roughly speaking, 0 indicates impossibility of the event and 1 indicates certainty.
- The higher the probability of an event, the more likely it is that the event will occur.

Basic concepts of probability

- Random Experiments
- Sample space
- Events
 - ① Mutually exclusive events (Disjoint events)
 - ② Equally likely events - equal chance to occur.
 - ③ Favourable events - the number of outcomes favourable to an event in an experiment is the number of outcomes which entail the happening of the event
 - ④ Exhaustive events - outcomes are said to be exhaustive when they include all possible outcomes.
 - ⑤ Independent events - if the occurrence or non-occurrence of an event does not affect the occurrence or non-occurrence of the other.

Examples - Applications Of Probability in Real Life

Probability is used to answer the following types of questions:

- What is the chance that it will rain tomorrow?
- What is the chance that a stock will go up in price?
- What is the chance that I will have a heart attack?
- What is the chance that I will live longer than 70 years?
- What is the likelihood that when rolling a pair of dice, I will roll doubles?
- What is the probability that I will win the lottery?
- What is the probability that I will become diabetic?

...Examples - Applications Of Probability in Real Life

- Forecasting the weather.
- Sports outcomes. Coaches use probability to decide the best possible strategy to pursue in a game.
- Card games and other games of chance.
- Insurance.
- Medical diagnosis.
 - This is one of the most noble applications of probability in real life.
 - How does your doctor know that your cough is just because of an infection and not because of something more serious?
- Election results.
- Lottery probability.

Random variable

- A real variable X whose value is determined by the outcome of a random experiment is called a random variable.
- A random variable, X , is a function that assigns a single, but variable, value to each element of a sample space.
- **Example:** A random experiment consists of two tosses of a coin. Consider the random variable which is the number of heads (0, 1, 2).

Outcome	Value of X
HH	2
HT	1
TH	1
TT	0

Discrete and Continuous Random Variables

- ① A discrete random variable:
 - has a countable number of possible values
 - has discrete jumps (or gaps) between successive values
 - has measurable probability associated with individual values
 - counts
- ② A continuous random variable:
 - has an uncountably infinite number of possible values
 - moves continuously from value to value
 - has no measurable probability associated with each value
 - measures (e.g.: height, weight, speed, value, duration, length)

Discrete Vs Continuous Probability Distribution

Discrete Probability Distribution	Continuous Probability Distribution
Bernouli	Normal
Binomial	Exponential
Poisson	Gamma
Hypergeometric	Logistic
Multinomial	Weibul
Geometric	Log-logistic
Negative Binomial	Log-normal
⋮	⋮

The Binomial Random Variable

- Consider a Bernoulli Process in which we have a sequence of n identical trials satisfying the following conditions:
 - ① Each trial has two possible outcomes, called success and failure. The two outcomes are mutually exclusive and exhaustive.
 - ② The probability of success, denoted by p , remains constant from trial to trial. The probability of failure is denoted by q , where $q = 1 - p$.
 - ③ The n trials are independent. That is, the outcome of any trial does not affect the outcomes of the other trials.
- A random variable, X , that counts the number of successes in n Bernoulli trials, where p is the probability of success in any given trial, is said to follow the binomial probability distribution with parameters n (number of trials) and p (probability of success). We call X the binomial random variable.

What is Probability Distribution?

The probability distribution of a random variable is a table that lists the possible values of the random variables and their associated probabilities.

- Example: Consider the different possible orderings of boy (B) and girl (G) in four sequential births. There are $2*2*2*2=16$ possibilities, so the sample space is:

x	$p(x)$
0	$\frac{1}{16}$
1	$\frac{4}{16}$
2	$\frac{6}{16}$
3	$\frac{4}{16}$
4	$\frac{1}{16}$

The Binomial Probability Distribution

What is Binomial Distribution?

- Binomial distribution is a common probability distribution that models the probability of obtaining one of two outcomes under a given number of parameters.
- It summarizes the number of trials when each trial has the same chance of attaining one specific outcome.
- The binomial distribution is frequently used to model the number of successes in a sample of size n drawn with replacement from a population of size N .
- If the sampling is carried out without replacement, the draws are not independent and so the resulting distribution is a hypergeometric distribution, not a binomial one.

...The Binomial Probability Distribution

The binomial probability distribution:

$$P(x) = \binom{n}{x} P^x (1 - p)^{n-x} = \frac{n!}{x!(n-x)!} P^x (1 - p)^{n-x}$$

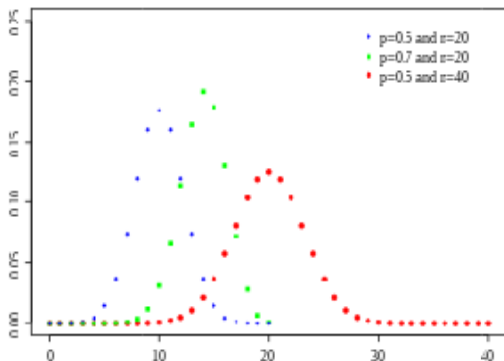
where

- p is the probability of success in a single trial
- $q = 1 - p$ = probability of failure on a single trial
- n is the number of trials
- x is the number of successes.

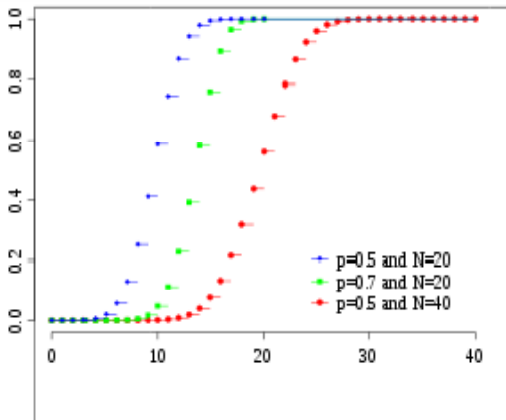
Binomial distribution

- ① Mean = $\mu = np$
- ② Variance = $\sigma^2 = npq$
- ③ Standard Deviation = $\sigma = \sqrt{npq}$
- ④ Moment Generating function (MGF) = $(q + pe^t)^n$
- ⑤ Characteristic function (CF) = $(q + pe^{it})^n$
- ⑥ Skewness = $\frac{q-p}{\sqrt{npq}}$
- ⑦ Kurtosis = $\frac{1-6pq}{npq}$

Binomial distribution - Probability mass function



Binomial distribution - Cumulative distribution function



Examples of the Binomial Distribution

- Example 1. A family has four children. Assuming the probability of a male birth is $\frac{1}{2}$, find the probability that there will be
 - 1 at least one boy
 - 2 at least one boy and one girl
- Solution:
- Let X and Y denote the number of boys and girls in the family, respectively.
- Given $P(x) = \frac{1}{2}$ and $P(y) = \frac{1}{2}$

$$P(\text{at least one boy}) = P(X \geq 1)$$

$$\begin{aligned}
 &= 1 - P(X = 0) = 1 - \left[\binom{4}{0} \left(\frac{1}{2}\right)^4 \right] \\
 &= 1 - \frac{1}{16} = \frac{15}{16}
 \end{aligned}$$

Examples of the Binomial Distribution

...Example 1

P(at least one boy and one girl)

$$\begin{aligned}
 &= P[(X \geq 1) \cap (Y \geq 1)] \\
 &= 1 - P[(X < 1) \cap (Y < 1)] \\
 &= 1 - [P(X = 0) + P(Y = 0)] \\
 &= 1 - \left[\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4\right] \\
 &= 1 - \left(\frac{1}{16} + \frac{1}{16}\right) \\
 &= \frac{7}{8}
 \end{aligned}$$

...Examples of the Binomial Distribution

- Example 2. In a research, rats are injected with a drug that inhibits body synthesis of protein. By the previous research, it was found that the probability of a rat dying from the drug before the experiment is over is 0.2. If 10 rats are used.
 - ① How many are expected to die before the experiment ends.
 - ② What is the probability that at least eight will survive/
- Solution:
- Let X be the random variable representing the number of rats that die in the experiment.
- Here, $P = 0.2, \Rightarrow q = 0.8, n = 10$

...Examples of the Binomial Distribution

...Example 2 Expected number of rats that die before the experiment =
 $10 \times 0.2 = 2$

$P(\text{at least eight will survive}) = P(\text{at most 2 will die})$

$$\begin{aligned}
 P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\
 &= \binom{10}{0} (0.2)^0 (0.8)^{10} + \binom{10}{1} (0.2)^1 (0.8)^9 + \binom{10}{2} (0.2)^2 (0.8)^8 \\
 &= 0.6778
 \end{aligned}$$

...Examples of the Binomial Distribution

Example 3: Assuming half of the population are consuming rice, so that the chance of an individual being a consumer is $\frac{1}{2}$ and assuming that 100 investigation take each 10 individuals to check whether they are consumers. How many investigators do we expect to report three or less were consumers.

Solution:

- Let X be the random variable representing the number of consumers of rice in 10 individuals.
- $n = 10, p = \frac{1}{2}, q = \frac{1}{2}$
- $P(X = x) = \binom{10}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} = \binom{10}{x} \left(\frac{1}{2}\right)^{10}$

...Examples of the Binomial Distribution

...Example 3

$$\begin{aligned}
 P(X \leq 3) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\
 &= \binom{10}{0} \left(\frac{1}{2}\right)^{10} + \binom{10}{1} \left(\frac{1}{2}\right)^{10} + \binom{10}{2} \left(\frac{1}{2}\right)^{10} + \binom{10}{3} \left(\frac{1}{2}\right)^{10} \\
 &= \left(\frac{1}{2}\right)^{10} (1 + 10 + 45 + 120) \\
 &= \frac{176}{1024} = 0.1718
 \end{aligned}$$

...Examples of the Binomial Distribution

Example 4: Number of Side Effects from Medications

- Medical professionals use the binomial distribution to model the probability that a certain number of patients will experience side effects as a result of taking new medications.
- For example, suppose it is known that 5% of adults who take a certain medication experience negative side effects.
- We can use a Binomial Distribution Calculator to find the probability that more than a certain number of patients in a random sample of 100 will experience negative side effects.
 - ① $P(X > 5 \text{ patients experience side effects}) = 0.38400$
 - ② $P(X > 10 \text{ patients experience side effects}) = 0.01147$
 - ③ $P(X > 15 \text{ patients experience side effects}) = 0.0004$

...Examples of the Binomial Distribution

Example 5: Number of Spam Emails per Day

- Email companies use the binomial distribution to model the probability that a certain number of spam emails land in an inbox per day.
- For example, suppose it is known that 4% of all emails are spam.
- If an account receives 20 emails in a given day, we can use a Binomial Distribution Calculator to find the probability that a certain number of those emails are spam:
 - 1 $P(X = 0 \text{ spam emails}) = 0.44200$
 - 2 $P(X = 1 \text{ spam email}) = 0.36834$
 - 3 $P(X = 2 \text{ spam emails}) = 0.14580$

The Poisson Probability Distribution

What is Poisson Distribution?

- The Poisson probability distribution is useful for determining the probability of a number of occurrences over a given period of time or within a given area or volume.
- That is, the Poisson random variable counts occurrences over a continuous interval of time or space.
- A Poisson distribution is a discrete probability distribution.
- It gives the probability of an event happening a certain number of times (k) within a given interval of time or space. The Poisson distribution has only one parameter, (λ), which is the mean number of events.

...The Poisson Probability Distribution

- The Poisson distribution represents the probability distribution of a certain number of events occurring in a fixed time interval.
- The events tend to have a constant mean rate.
- Poisson distribution is further used to determine how many times an event is likely to occur within a given time period.
- The range of Poisson distribution starts at zero, and it goes until infinity.
- In Poisson distribution, the rate at which the events occur must be constant, and the occurrence of one event must not affect the occurrence of any other event, i.e., the events should occur independently.

Examples of Poisson Distribution

- 1 Number of Network Failures per Week
- 2 Number of Bankruptcies Filed per Month
- 3 Number of Website Visitors per Hour
- 4 Number of Arrivals at a Restaurant
- 5 Number of Calls per Hour at a Call Center
- 6 Number of Books Sold per Week
- 7 Average Number of Storms in a City
- 8 Number of Emergency Calls Received by a Hospital Every Minute

...The Poisson Probability Distribution

A discrete random variable X is said to have a Poisson distribution, with parameter $\lambda > 0$, if it has a probability mass function given by:

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

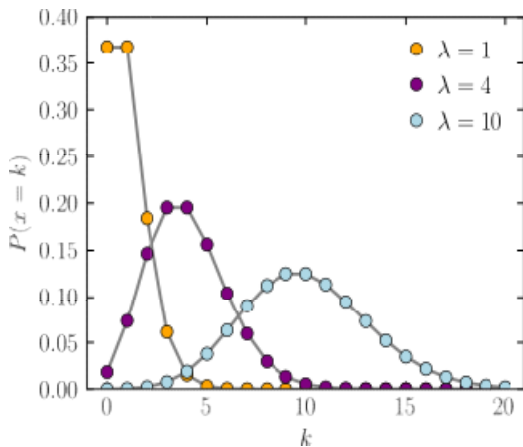
for $x = 1, 2, 3, \dots$

- The events are independent.
- The average number of successes in the given period of time alone can occur.
- No two events can occur at the same time.
- The Poisson distribution is limited when the number of trials n is indefinitely large.

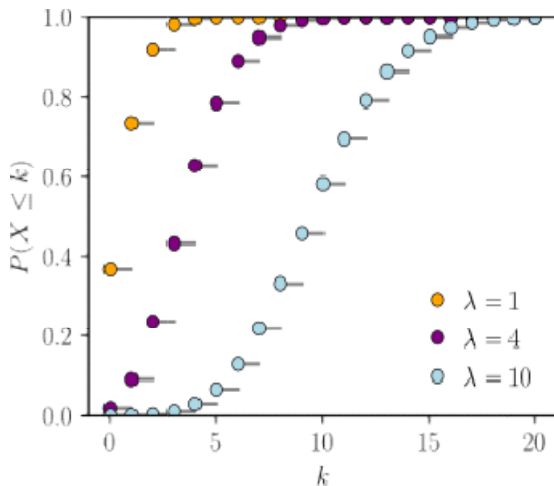
Poisson distribution

- 1 Mean = λ
- 2 Variance = λ
- 3 Standard Deviation = $\sqrt{\lambda}$
- 4 Moment Generating function (MGF) = $\exp[\lambda(e^t - 1)]$
- 5 Characteristic function (CF) = $\exp[\lambda(e^{it} - 1)]$
- 6 Skewness = $\frac{1}{\sqrt{\lambda}}$
- 7 Kurtosis = $\frac{1}{\lambda}$

Poisson distribution - Probability mass function



Poisson distribution - Cumulative distribution function



Poisson distribution is limiting case of binomial distribution

- ① The number of trials is indefinitely large i.e, $n \rightarrow \infty$
- ② p the probability of success in each trial is very small i.e, $p \rightarrow 0$
- ③ $np(= \lambda)$ is finite and $p = \frac{\lambda}{n}$
- ④ $q = 1 - \frac{\lambda}{n}$

Computations of Poisson Probabilities

Example 1: A life insurance company insures the lives of 5000 men of age 42 years. The probability of a 42 year old man dying in a given year is 0.001. What is the probability that the company will have to pay 4 claims in a given year.

Solution:

- Total number of men insured = $n = 5000$
- Probability = $p = 0.001$
- $\lambda = np = 5000 \times 0.001 = 5$.
- Thus,

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-5} 5^x}{x!}$$

$$P(X = 4) = \frac{e^{-5} 5^4}{4!} = 0.1755$$

Computations of Poisson Probabilities

Example 2: A travel company has two cars for hiring. The demand for a car on each day is distributed as poisson variate, with mean 1.5. Calculate the proportion of days on which

- ① Neither cars were used
 - ② Some demand is refused
- Solution
 - Let X be random variable representing the number of demands for cars.
 - $\lambda = 1.5$

$$P(X = x) = P(x \text{ demands in a day}) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-1.5} (1.5)^x}{x!}$$

Computations of Poisson Probabilities

...Example 2

- 1 The proportion of days in which neither car is used

$$P(X = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} = 0.2231$$

- 2 The proportion of days on which some demand is refused. The demand is refused when X goes more than 2

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right] \\ &= 1 - e^{-1.5} \left[1 + 1.5 + \frac{(1.5)^2}{2!} \right] = 0.19126 \end{aligned}$$

Computations of Poisson Probabilities

Example 3: The probability of an individual suffering a bad reaction from an injection of a certain antibiotic is 0.001. Out of 2000 individuals, find the probability that

- ① Exactly 3 suffer
- ② More than 2 suffer from bad reaction

Solution:

- Let X denote the number of individuals suffering from bad reaction

$$P = 0.001 \text{ and } n = 2000$$

- Since n is a large and probability of bad reaction is small, we can assume X as a Poisson variate with $\lambda = np = 2000 \times 0.001 = 2$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2} (2)^x}{x!}$$

Computations of Poisson Probabilities

...Example 3

$$P(\text{exactly 3 suffer}) = P(X = 3) = \frac{e^{-2}(2)^3}{3!} = 0.180447$$

$$\begin{aligned}
 P(\text{more than 2 suffer}) &= 1 - P(X \leq 2) \\
 &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\
 &= 1 - \left[\frac{e^{-2}(2)^0}{0!} + \frac{e^{-2}(2)^1}{1!} + \frac{e^{-2}(2)^2}{2!} \right] \\
 &= 1 - e^{-2}[1 + 2 + 2] \\
 &= 1 - 0.6767 = 0.3233
 \end{aligned}$$

Computations of Poisson Probabilities

Example 4: An insurance company found that only 0.01% of the population is involved in road accident in a particular area. If 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients will be involved in such an accident next year.

Solution:

- $p = 0.01\% = 0.0001$
- $n = 1000$
- $\lambda = np = 0.000 \times 1000 = 0.1$
- Let X be a random variable of clients involved in accident

$$P(X = x) = \frac{e^{-\lambda}(\lambda)^x}{x!} = \frac{e^{-0.1}(0.1)^x}{x!}$$

Computations of Poisson Probabilities

...Example 4

$$\begin{aligned}
 P(\text{not more than two}) &= P(X \leq 2) \\
 &= P(X = 0) + P(X = 1) + P(X = 2) \\
 &= \frac{e^{-0.1}(0.1)^0}{0!} + \frac{e^{-0.1}(0.1)^1}{1!} + \frac{e^{-0.1}(0.1)^2}{2!} \\
 &= e^{-0.1} \left[1 + 0.1 + \frac{(0.1)^2}{2} \right] = 0.9998
 \end{aligned}$$

The Normal Probability Distribution

What Is a Normal Distribution?

- Normal distribution, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.
- In graphical form, the normal distribution appears as a "bell curve"

...The Normal Probability Distribution

What are the 5 properties of a normal distribution?

- It is symmetric. A normal distribution comes with a perfectly symmetrical shape.
- The mean, median, and mode are equal.
- The middle point of a normal distribution is the point with the maximum frequency, which means that it possesses the most observations of the variable.
- Empirical rule.
- Skewness and kurtosis.

...The Normal Probability Distribution

In statistics, a normal distribution or Gaussian distribution is a type of continuous probability distribution for a real-valued random variable. The general form of its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

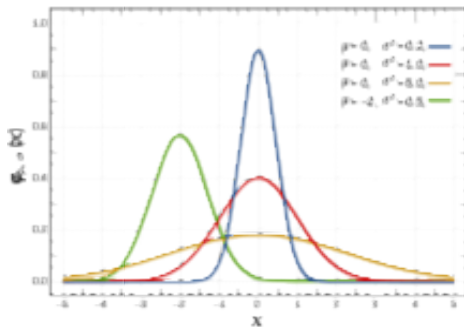
for $x = 1, 2, 3, \dots$

- The parameter μ is the mean or expectation of the distribution (and also its median and mode), while the parameter σ is its standard deviation.
- The variance of the distribution is σ^2 .
- A random variable with a Gaussian distribution is said to be normally distributed, and is called a normal deviate.

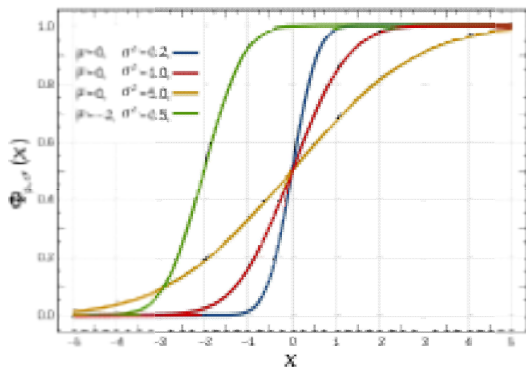
Normal distribution

- ① Mean = μ
- ② Variance = σ^2
- ③ Standard Deviation = σ
- ④ Moment Generating function (MGF) = $\exp[\mu t + \sigma^2 \frac{t^2}{2}]$
- ⑤ Characteristic function (CF) = $\exp[i\mu t + \sigma^2 \frac{t^2}{2}]$
- ⑥ Skewness = 0
- ⑦ Kurtosis = 0

Normal distribution - Probability mass function



Normal distribution - Cumulative distribution function



Standard Normal Distribution

What is Standard Normal distribution?

- If X is a random variable following normal distribution with parameter μ and σ , then $Z = \frac{X-\mu}{\sigma}$ is called the standard normal variate and the probability density function of the standard variate Z is given by

$$\phi(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < x < \infty$$

Examples of Normal Distribution

- 1 Birthweight of Babies
- 2 Height of Babies
- 3 Shoe Sizes
- 4 Blood Pressure
- 5 Students mark

Computations of Normal Probabilities

Example 1. Let $X \sim N(\mu = 160, \sigma^2 = 30^2)$

Compute $P(100 \leq X \leq 180) = ?$

$$\begin{aligned}
 P(100 \leq X \leq 180) &= P\left(\frac{100 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{180 - \mu}{\sigma}\right) \\
 &= P\left(\frac{100 - 160}{30} \leq Z \leq \frac{180 - 160}{30}\right) \\
 &= P(-2 \leq Z \leq 0.6667) \\
 &= 0.4772 + 0.2475 \\
 &= 0.7247
 \end{aligned}$$

Computations of Normal Probabilities

Example 2. Let $X \sim N(\mu = 127, \sigma^2 = 22^2)$

Compute $P(X \leq 150) = ?$

$$\begin{aligned} P(X \leq 150) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{150 - \mu}{\sigma}\right) \\ &= P\left(Z \leq \frac{150 - 127}{22}\right) \\ &= P(Z \leq 1.045) \\ &= 0.5 + 0.3520 \\ &= 0.8520 \end{aligned}$$

Computations of Normal Probabilities

Example 3. Suppose the height of men of a certain country are normally distributed with average 68 and $SD = 2.5$. Find the percentage of men who are between $a = 66$ and $b = 71$ inches in height.

Solution:

$$\begin{aligned}
 P(66 \leq X \leq 71) &= P\left(\frac{66 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{71 - \mu}{\sigma}\right) \\
 &= P\left(\frac{66 - 68}{2.5} \leq Z \leq \frac{71 - 68}{2.5}\right) \\
 &= P(-0.8 \leq Z \leq 1.20) \\
 &= P(0 \leq Z \leq 0.8) + P(0 \leq Z \leq 1.20) \\
 &= 0.2881 + 0.3849 \\
 &= 0.673
 \end{aligned}$$

Approximately 67.3% men are between 66 and 71 inches in height.

Computations of Normal Probabilities

Example 4. If X is normally distributed with $\mu = 12$ and $\sigma = 4$. Find the probability of,

- ① $X \geq 20$
- ② $X \leq 20$
- ③ $0 \leq X \leq 12$

Solution:

$$\begin{aligned}
 P(X \geq 20) &= P\left(\frac{X - \mu}{\sigma} \geq \frac{20 - \mu}{\sigma}\right) \\
 &= P\left(Z \geq \frac{20 - 12}{4}\right) \\
 &= P(Z \geq 2) \\
 &= 0.5 - 0.4772 \\
 &= 0.0228
 \end{aligned}$$

Computations of Normal Probabilities

...Example 4 Solution:

$$P(X \leq 20) = 1 - P(X \geq 20) = 1 - 0.0228 = 0.9772$$

$$\begin{aligned} P(0 \leq X \leq 12) &= P\left(\frac{0 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{12 - \mu}{\sigma}\right) \\ &= P\left(\frac{0 - 12}{4} \leq Z \leq \frac{12 - 12}{4}\right) \\ &= P(-3 \leq Z \leq 0) \\ &= P(0 \leq Z \leq 3) \\ &= 0.4987 \end{aligned}$$

Computations of Normal Probabilities

Exercise. If X is normally distributed with $\mu = 30$ and $\sigma = 5$. Find the probability of,

- ① $P(26 \leq X \leq 40) = ?$
- ② $P(X \geq 45) = ?$

What is statistical data?

When census data cannot be collected, statisticians collect sample data by developing specific experiment designs and survey samples.

- How do we get representative samples?
- How do we get optimal sample?

- What is Sampling?
- Why we need it?
- When we need it?
- How do we get it?
- How much we need it?

Definition of some basic terms

- A Population consists of the set of all measurements/elements under study
- Population: is the totality of elements or units under study.
- A survey is any activity that collects information in an organized and methodical manner about characteristics of interest from some or all units of a population using well-defined concepts, methods and procedures, and compiles such information into a useful summary form (Statistics Canada, 2010)
- Census Surveys: data are collected for all units in the population
- Sample Surveys: data are collected for only a fraction (typically a very small fraction) of units of the population
- Sampling frame: A complete list of all the units of the population
- Sample: is the part of the population.
- Sampling: is the techniques of selecting representative sample from the population

Parameter Vs Statistic

A parameter is a number describing a whole population (e.g., population mean), while a statistic is a number describing a sample (e.g., sample mean).

Sample			Population	
n	←	size	→	N
$\bar{X}$	←	mean	→	μ
S^2	←	variance	→	σ^2
s	←	st.dev	→	σ
$\hat{p}$	←	Proportion	→	P

Reasons for Sampling

- Using sample saves time, cost/finance and human resources
- It prevents destruction
- It provides higher level of accuracy
- It may be the only way of undertaking the study

What are the limitations of sampling?

Impossibility of sampling.

- Chances of bias. The serious limitation of the sampling method is that it involves biased selection and thereby leads us to draw erroneous conclusions.
- Difficulties in selecting a truly representative sample.
- Inadequate knowledge in the subject.
- Changeability of units.
- Impossibility of sampling.

What factors do we consider when sampling?

- The reasons for and objectives of sampling.
- The relationship between accuracy and precision.
- The reliability of estimates with varying sample size.
- The determination of safe sample sizes for surveys.
- The variability of data.

Types of Sampling Methods

- Probability sampling involves random selection, allowing you to make strong statistical inferences about the whole group.
- Non-probability sampling involves non-random selection based on convenience or other criteria, allowing you to easily collect data.

Probability (random) Sampling

- Simple Random Sampling
- Systematic Random Sampling
- Stratified Random Sampling
- Cluster Random Sampling
- Multistage Random Sampling
- Multiphase Random Sampling
- Matched Random Sampling
- Panel Random Sampling

....Probability (random) Sampling

Probability (random) Sampling - the selection of the sample is purely based on chance. Every unit of the population has a known non-zero probability of to be included in the sample.

- Simple random sampling: every unit of the population is given an equal chance of being selected. The sample can be drawn using lottery method or table of random numbers
- Systemic sampling: in systemic sampling, we start at a random point in the sampling frame, and from this point selected every kth, say, value in the frame to formulate the sample.
- Stratified sampling: in stratified sampling, the population is partitioned into two or more subpopulation called strata, and from each stratum a desired sample size is selected at random.

$$n_h = N_h \frac{n}{N}$$

- Cluster sampling: in cluster sampling, a random sample of the strata is selected and then samples from these selected strata are obtained.

....Probability (random) Sampling

- Multistage Sampling

- Complex form of cluster sampling in which two or more levels of units are embedded one in the other.
- First stage, random number of districts chosen in all states.
- Followed by random number of villages.
- Then third stage units will be houses.
- All ultimate units (houses, for instance) selected at last step are surveyed.

....Probability (random) Sampling

• ...Multistage Sampling

- This technique, is essentially the process of taking random samples of preceding random samples.
- Not as effective as true random sampling, but probably solves more of the problems inherent to random sampling.
- An effective strategy because it banks on multiple randomizations. As such, extremely useful.
- Multistage sampling used frequently when a complete list of all members of the population not exists and is inappropriate.
- Moreover, by avoiding the use of all sample units in all selected clusters, multistage sampling avoids the large, and perhaps unnecessary, costs associated with traditional cluster sampling.

....Probability (random) Sampling

- Multiphase Sampling

- Part of the information collected from whole sample and part from subsample.
- In Tb survey MT in all cases Phase I
- X Ray chest in MT +ve cases Phase II
- Sputum examination in X Ray +ve cases - Phase III
- Survey by such procedure is less costly, less laborious and more purposeful

....Probability (random) Sampling

- Matched Random Sampling

- A method of assigning participants to groups in which pairs of participants are first matched on some characteristic and then individually assigned randomly to groups.
- Two samples in which the members are clearly paired, or are matched explicitly by the researcher.
- For example, IQ measurements or pairs of identical twins.
- Those samples in which the same attribute, or variable, is measured twice on each subject, under different circumstances. Commonly called repeated measures.
- Examples include the times of a group of athletes for 1500m before and after a week of special training; the milk yields of cows before and after being fed a particular diet.

....Probability (random) Sampling

- Panel Sampling

- Method of first selecting a group of participants through a random sampling method and then asking that group for the same information again several times over a period of time.
- Therefore, each participant is given same survey or interview at two or more time points; each period of data collection called a "wave".
- This sampling methodology often chosen for large scale or nation-wide studies in order to gauge changes in the population with regard to any number of variables from chronic illness to job stress to weekly food expenditures.
- Panel sampling can also be used to inform researchers about within-person health changes due to age or help explain changes in continuous dependent variables such as spousal interaction.
- There have been several proposed methods of analyzing panel sample data, including growth curves.

Non-probability Sampling

Non-probability Sampling - the sample is not based on chance. It is based on personal judgment.

- Quota Sampling
- Judgment or purposive Sampling
- Convenience Sampling
- Snowball sampling
 - Snowball sampling or chain-referral sampling is defined as a non-probability sampling technique in which the samples have traits that are rare to find.
 - Snowball sampling is where research participants recruit other participants for a test or study.
 - As sample members are not selected from a sampling frame, snowball samples are subject to numerous biases. For example, people who have many friends are more likely to be recruited into the sample. When virtual social networks are used, then this technique is called virtual snowball sampling

What do you need to consider before deciding the right sampling technique?

How to Choose the Best Sampling Method

- List the research goals (usually some combination of accuracy, precision, and/or cost).
- Identify potential sampling methods that might effectively achieve those goals.
- Test the ability of each method to achieve each goal.

What are the factors that could affect the choice of sampling method?

- Choices in sample design are influenced by many factors, including the desired level of precision and detail of the information to be produced, the availability of appropriate sampling frames, the availability of suitable auxiliary variables for stratification and sample selection.

Sample Size Calculation

- Many of the studies published in several journals have less than required sample size and hence less power.
- Many articles have been published in existing literature explaining the methods of calculation of sample size but still a lot of confusion exists.
- It is very important to understand that method of sample size calculation is different for different study designs and one blanket formula for sample size calculation cannot be used for all study designs.

...Sample Size Calculation

Why we need to find optimal sample size?

- The main aim of a sample size calculation is to determine the number of participants needed to detect a clinically relevant treatment effect.
- Studies should be designed to include a sufficient number of participants to adequately address the research question.
- Studies that have either an inadequate number of participants or an excessively large number of participants are both wasteful in terms of participant and investigator time, resources to conduct the assessments, analytic efforts and so on.

...Sample Size Calculation

Why sample size is important?

- The power of a statistical test is the probability that a test will reject the null hypothesis when the null hypothesis is false.
- That is, power reflects the probability of not committing a type II error.
- The two major factors affecting the power of a study are the sample size and the effect size
- The larger the sample size is the smaller the effect size that can be detected.
- The reverse is also true; small sample sizes can detect large effect sizes.
- While researchers generally have a strong idea of the effect size in their planned study it is in determining an appropriate sample size that often leads to an underpowered study.

...Sample Size Calculation

Why sample size is important?

- The two major factors affecting the power of a study are the sample size and the effect size
- A study should only be undertaken once there is a realistic chance that the study will yield useful information
- A study that has a sample size which is too small may produce inconclusive results and could also be considered unethical by exposing human subjects or lab animals to needless risk
- A study that is too large will waste scarce resources and could expose more participants than necessary to any related risk
- Thus an appropriate determination of the sample size used in a study is a crucial step in the design of a study

...Sample Size Calculation

Before directly heading to calculate a sample size, you need to determine a few things about the target population and the sample you need?

- Population Size: How many total people fit your demographic?
- Margin of Error (Confidence Interval): No sample will be perfect, so you need to decide how much error to allow. The confidence interval determines how much higher or lower than the population mean you are willing to let your sample mean fall.
- Confidence Level: How confident do you want to be that the actual mean falls within your confidence interval? The most common confidence intervals are 90% confident, 95% confident, and 99% confident.
- Standard of Deviation: How much variance do you expect in your responses?

What factors affect the sample size?

- The factors affecting sample sizes are study design, method of sampling, and outcome measures effect size, standard deviation, study power, and significance level.
- How does variation affect sample size?
 - As a sample size increases, sample variance (variation between observations) increases but the variance of the sample mean (standard error) decreases and hence precision increases.

What is the appropriate Sample Size for my study?

- The answer depends on number of factors, including the purpose of your study, population of your study, population size, risk of selecting a bad sample and the allowable sampling error.
- In addition to the purpose of the study and population size, three criteria usually will need to be specified to determine the appropriate sample size (Miaoulis and Michener, 1976)
 - Level of precision
 - Level of confidence or risk
 - Degree of variability in the attributes being measured.

Level of Precision

- A measure of how close an estimate is to the true value of a population parameter.
- It is the range in which the true value of the population is estimated to be
- This range is often expressed in percentage points.
- It may be expressed in absolute terms or relative to the estimate $\pm 3\%$, $\pm 5\%$, $\pm 10\%$...

α and Confidence level

- α : The significance level of a test the probability of rejecting the null hypothesis when it is true (or the probability of making a Type I error)
- Confidence level: The probability that an estimate of a population parameter is within certain specified limits of the true value; commonly denoted by $1 - \alpha$
- Also known as risk level
- Based on CLT, which means when a population is repeatedly sampled, the average value of the attribute obtained by those samples is equal to the true population value.
- This is expressed in percentage points, for example 95%

Degree of Variability

- It refers to the distribution of attributes in the population
- The more heterogeneous a population, the larger the sample size required to obtain a given level of precision
- The less variable (more homogeneous) a population, the smaller the sample size.

β and Power

- β : The probability of failing to reject the null hypothesis when it is false (or the probability of making a Type II error)
- Power: The probability of correctly rejecting the null hypothesis when it is false; commonly denoted by $1 - \beta$

Steps in Estimating Sample Size

- Identify major variables (Qualitative or Quantitative)
- Determine type of estimate (% , mean, ratio, proportion,...)
- Indicate expected frequency of factor of interest
- Decide on desired precision of the estimate
- Decide on acceptable risk that estimate will fall outside its real population value
- Adjust for population size
- Adjust for estimated designed effects
- Adjust for expected response rate

What is the design effect in sampling?

- The design effect (denoted as $deff$) is defined as the ratio of the variance of an estimate under a sampling plan to the variance of the same estimate from a simple random sample with same number of observation units.
- The sampling plan could be a stratified sampling or other complex sample designs.

What is design effect in sample size calculation?

- A design effect (DEFF) is an adjustment made to find a survey sample size, due to a sampling method (e.g. cluster sampling, respondent driven sampling, or stratified sampling) resulting in larger sample sizes (or wider confidence intervals) than you would expect with simple random sampling (SRS).

Is sample size part of study design?

- Calculation of exact sample size is an important part of research design.
- It is very important to understand that different study design need different method of sample size calculation and one formula cannot be used in all designs.

How can design and sample size affect results?

- The use of sample size calculation directly influences research findings.
- Very small samples undermine the internal and external validity of a study.
- Very large samples tend to transform small differences into statistically significant differences - even when they are clinically insignificant.

Study Designs

- Cross-sectional studies
- Case control studies
- Cohort studies

Sample Size Calculation for Cross-sectional studies/survey

- Cross sectional studies or cross sectional survey are done to estimate a population parameter like prevalence of some disease in a community or finding the average value of some quantitative variable in a population.
- Sample size formula for qualitative variable and quantities variable are different.

...Sample Size Calculation for Cross-sectional studies/survey

For Qualitative variable

- Suppose an epidemiologist want to know proportion of children who are hypertensive in a population then this formula should be used as proportion is a qualitative variable.

$$n_0 = \frac{Z_{\alpha/2}^2 P(1-P)}{E^2}$$

$$n = \frac{n_0}{1 + \frac{n_0}{N}} \text{ for finite population and } \frac{n_0}{N} > 5\%$$

where

- Z is the value from the standard normal distribution reflecting the confidence level that will be used (e.g., $Z = 1.96$ for 95%)
- E is the desired margin of error/Absolute error or precision Has to be decided by researcher.
- P is the proportion of successes in the population. Here we are planning a study to generate a 95% confidence interval for the unknown population proportion, p.

Example 1

- Let us assume that a researcher wants to estimate proportion of patients having hypertension in paediatric age group in a city.
- According to previously published studies actual number of hypertensives may not be more than 15%.
- The researcher wants to calculate this sample size with the precision/absolute error of 5% and at Type I error of 5%.

$$n = \frac{Z_{\alpha/2}^2 P(1-P)}{E^2} = \frac{1.96^2 \times 0.15(1-0.15)}{0.05^2} = 196$$

- So for this cross sectional study researcher has to take at least 196 subjects.
- If the researcher want to increase the error (decrease the precision) then denominator will increase and hence sample size will decrease.

...Sample Size Calculation for Cross-sectional studies/survey

For Quantitative variable

- Suppose the same researcher is interested in knowing average systolic blood pressure of children of the same city. The researcher should consider blood pressure is a quantitative variable.

$$n = \frac{Z_{\alpha/2}^2 sd^2}{E^2}$$

where

- Z is the value from the standard normal distribution reflecting the confidence level that will be used (e.g., $Z = 1.96$ for 95%)
- E is the desired margin of error/Absolute error or precision Has to be decided by researcher.
- SD = Standard deviation of variable. Value of standard deviation can be taken from previously done study or through pilot study.

Example 2

- If the researcher is interested in knowing the average systolic blood pressure in pediatric age group of that city at 5% of Type I error and precision of 5 mmHg of either side (more or less than mean systolic BP) and standard deviation, based on previously done studies, is 25 mmHg then formula for sample size calculation will be

$$n = \frac{1.96^2 \times 25^2}{5^2} = 96$$

- So researcher will have to take the blood pressure of at least 96 children to know average systolic blood pressure in paediatric age group.

Sample Size Calculation for case control studies

- In case control studies cases (the group with disease/ condition under consideration) are compared with controls (the group without disease/condition under consideration) regarding exposure to the risk factor under question.
- The formula for sample size calculation for this design also depends on the type of variable (qualitative or quantitative). Here formula for independent case control study.

For qualitative variable

- Suppose a researcher want to see the link between childhood sexual abuses with psychiatric disorder in adulthood.
- He will take a sample of adult persons with psychiatric disorder and will take another sample of normal adults having no psychiatric disorders.
- He will then go retrospectively to see history of childhood sexual abuse in both groups.
- Exposure to both groups will be compared and odds ratio will be

...Sample Size Calculation for case control studies

- Here number of people exposed to childhood sexual abuse is qualitative variable hence this formula will be used for such type of design

$$n = \frac{(r+1)(p^*)(1-p^*)(Z_{\beta}+Z_{\alpha})^2}{r(p_1-p_2)^2}, \text{ where}$$

- r = Ratio of control to cases, 1 for equal number of case and control
- p^* = Average proportion exposed = (proportion of exposed cases + proportion of control exposed)/2
- Z_{β} = Standard normal variate for power = for 80% power it is 0.84 and for 90% value is 1.28. Researcher has to select power for the study.
- $Z_{\alpha/2}$ = Standard normal variate for level of significance as mentioned in previous section.
- $p_1 - p_2$ = Effect size or different in proportion expected based on previous studies. p_1 is proportion in cases and p_2 is proportion in control.

...Sample Size Calculation for case control studies

- So if the researcher wants to calculate sample size for the above mentioned case control study to know link between childhood sexual abuse with psychiatric disorder in adulthood and he wants to fix power of study at 80% and assuming expected proportions in case group and control group are 0.35 and 0.20 respectively, and he wants to have equal number cases and control; then the sample size per group will be

$$n = \frac{2(p^*)(1-p^*)(0.84+1.96)^2}{1(0.35-0.20)^2}, \text{ where}$$

- $p^* = \text{Average proportion exposed} = \frac{0.35+0.20}{2} = 0.275$

$$n = \frac{2(0.275)(1-0.275)(0.84+1.96)^2}{1(0.35-0.20)^2} = 138.9 \approx 139$$

- So the researcher has to recruit at least 139 subjects in cases and equal number in control as he wants to have equal number in both.

...Sample Size Calculation for case control studies

For Quantitative variables

- Suppose a researcher wants to see the association between birth weight and diabetes in adulthood.
- The birth weight being a quantitative data, the researcher will select one group i.e. cases that will be diabetic adults and other group i.e. control will be nondiabetic adults.
- Both groups will be traced back for data regarding childhood weight.
- The formula for sample size calculation is

$$n = \frac{(r+1)sd^2(Z_{\beta}+Z_{\alpha/2})^2}{r \times d^2}, \text{ where}$$

- sd = Standard deviation = researcher can take value from previously published studies; d = Expected mean difference between case and control (may be based on previously published studies.); r , Z_{β} , $Z_{\alpha/2}$ are already explained in previous sections.

...Sample Size Calculation for case control studies

...For Quantitative variables

- For example, if researcher think that difference in mean weight between case and control may be around 250 gm and SD is 1 Kg then considering equal number of cases and control and 80% power the sample size will be

$$n = \frac{2 \times 1^2 (0.84 + 1.96)^2}{0.25^2} = 250.88 \approx 251$$

- Hence, researcher has to take 251 subjects in each group (case and control).

Sample Size Calculation for cohort studies

- In cohort studies healthy subjects with or without exposure to some risk factor are observed over a time period to see the event rate in both groups.
- If a researcher wants to see the impact of weight training exercise on cardiovascular mortality then he will select two groups, one consisting of subjects who do exercise and another consisting of those who don't do.
- These groups will be followed up for a specific time period to see cardiovascular mortality in both groups.
- At the end of the study period both groups will be compared for cardiovascular mortality.

...Sample Size Calculation for cohort studies

The formula for sample size is

$$n = \frac{\left[Z_{\alpha} \sqrt{\left(1 + \frac{1}{m}\right) p^* (1 - p^*)} + Z_{\beta} \sqrt{p_1 (1 - p_1) / m + p_2 (1 - p_2)} \right]^2}{(p_1 - p_2)^2}$$

- Z_{α} = Standard normal variate for level of significance
- m = Number of control subject per experimental subject
- Z_{β} = Standard normal variate for power or type 2 error as explained in earlier section
- p_1 = Probability of events in control group
- p_2 = Probability of events in experimental group p

$$p^* = \frac{p_2 + m p_1}{m + 1}$$

...Sample Size Calculation for cohort studies

- For example, so suppose the researcher wants to see the impact of weight training exercise on cardiovascular mortality and according to previous studies proportion of cardiovascular death in case may be around 20% and in control it can be around 40% hence sample size calculation for 5% of significant level and 80% power with equal number of case and control will be

$$n = \frac{\left[1.96\sqrt{(1+\frac{1}{1})0.30(1-0.30)} + 0.84\sqrt{0.40(1-0.40)/1+0.20(1-0.20)} \right]^2}{(0.40-0.20)^2} = 59.41 \approx 59$$

- Thus, the researcher has to take 59 samples in each group.

Sample Size Calculation for comparison between two groups for quantitative data

- When the variable is quantitative data like blood pressure, weight, height, etc., then the following formula can be used for calculation of sample size for comparison between two groups.

$$n = \frac{2 \times sd^2 (Z_{\alpha/2} + Z_{\beta})^2}{E^2}$$

- sd = standard deviation from previous studies or pilot study
- $Z_{\alpha/2} = Z_{0.05/2} = Z_{0.025} = 1.96$ (From Z table) at type 1 error of 5%
- $Z_{\beta} = Z_{0.20} = 0.842$ (From Z table) at 80% power
- d = effect size = difference between mean values

...Sample Size Calculation for comparison between two groups for quantitative data

- For example, suppose a researcher wants to see the effect of a potential antihypertensive drug and He wants to compare the new drug with placebo.
- Researcher thinks that if this new drug reduces this blood pressure by 10 mmHg as compared to placebo then it should be considered as clinically significant.
- Let us assume standard deviation found in previously done studies was 25 mmHg.
- Suppose the researcher selects the level of significance at 5% and the power of study at 80%., and he thinks suitable statistical test in this condition will be two tailed unpaired t test. The effect size in this condition is 10 mmHg. Hence sample size will be

$$n = \frac{2 \times 25^2 (1.96 + 0.84)^2}{10^2} = 98$$

- So in this case the researcher needs 98 subjects per group.

Sample Size Calculation for comparison between two groups for qualitative data

- When the endpoint of a clinical intervention study is qualitative like alive/dead, diseased/non diseased, male/ female etc., then the following formula can be used for sample size calculation for comparison between two groups.
- Suppose the researcher is interested in knowing protective effect of a drug on mortality in patients of myocardial infarction.
- He selected two groups of patients of myocardial infarction one group was given that drug and another group was given placebo.
- The both groups were kept under observation and at the end of study death in both groups were compared.

...Sample Size Calculation for comparison between two groups for qualitative data

- For sample size of this type of study below mentioned formula can be used.

$$n = \frac{2 \times [Z_{\alpha/2} + Z_{\beta}]^2 P(1-P)}{[p_1 - p_2]^2}$$

- $p_1 - p_2$ = Difference in proportion of events in two groups
- P = Pooled prevalence = [prevalence in case group (p_1) + prevalence in control group (p_2)]/2
- In above example, let us assume that previous study says that 20% of patient of myocardial infarction die within a specified time.

...Sample Size Calculation for comparison between two groups for qualitative data

- The researcher feels that if the drug being tested increases survival to 30% then the finding can be considered as clinically significant.
- Effect size will be difference between proportions. $0.2 - 0.3 = -0.1$.
- At 5% of significance level and 80% power sample size will be
- Pooled prevalence $= (0.20 + 0.30)/2 = 0.25$

$$n = \frac{2 \times [1.96 + 0.84]^2 \times 0.25(1 - 0.25)}{[-0.1]^2} = 294$$

- Thus, the researcher needs 294 subjects per group.

Summary for Sample Size Calculation

- Determining the appropriate design of a study is more important than the statistical analysis; a poorly designed study can never be salvaged, whereas a poorly analyzed study can be re-analyzed.
- A critical component in study design is the determination of the appropriate sample size.
- The sample size must be large enough to adequately answer the research question, yet not too large so as to involve too many patients when fewer would have sufficed.
- The determination of the appropriate sample size involves statistical criteria as well as clinical or practical considerations.
- Sample size determination involves teamwork; biostatisticians must work closely with clinical investigators to determine the sample size that will address the research question of interest with adequate precision or power to produce results that are clinically meaningful.

...Summary

The following table summarizes the sample size formulas for each scenario described here. The formulas are organized by the proposed analysis, a confidence interval estimate or a test of hypothesis.

Situation	To Estimate CI	To conduct HT
One Sample	$n = \frac{(Z_{\alpha/2}\sigma)^2}{E^2}$, Continuous	$n = \left[\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\frac{ \mu_1 - \mu_2 }{\sigma}} \right]^2$
Two Indep	$n_i = 2 \frac{(Z_{\alpha/2}\sigma)^2}{E^2}$	$n_i = 2 \left[\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\frac{ \mu_1 - \mu_2 }{\sigma}} \right]^2$
Two pairs	$n = \frac{(Z_{\alpha/2}\sigma_d)^2}{E^2}$	$n = \left[\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\frac{\mu_d}{\sigma_d}} \right]^2$
One Sample	$n = p(1-p) \frac{(Z_{\alpha/2})^2}{E^2}$, Dichotomous	$n = \left[\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\frac{p_1 - p_0}{\sqrt{p_1(1-p_1)}}} \right]^2$
Two Indep.	$n = [p_1(1-p_1) + p_2(1-p_2)] \frac{(Z_{\alpha/2})^2}{E^2}$	$n = \left[\frac{Z_{1-\alpha/2} + Z_{1-\beta}}{\frac{ p_1 - p_2 }{\sqrt{p(1-p)}}} \right]^2$

...Summary

- Estimating a population proportion with specified absolute precision

$$n = \frac{Z_{1-\alpha/2}P(1-P)}{e^2}$$

- Estimating a population proportion with specified relative precision

$$n = \frac{Z_{1-\alpha/2}(1-P)}{e^2P}$$

- Hypothesis tests for a population proportion

- For a one-sided test

$$n = \frac{\left[Z_{1-\alpha} \sqrt{P_0(1-P_0)} + Z_{1-\beta} \sqrt{P_a(1-P_a)} \right]^2}{[P_0 - P_a]^2}$$

- For a two-sided test

$$n = \frac{\left[Z_{1-\alpha/2} \sqrt{P_0(1-P_0)} + Z_{1-\beta} \sqrt{P_a(1-P_a)} \right]^2}{[P_0 - P_a]^2}$$

...Summary

- Estimating the difference between two population proportions with specified absolute precision

$$n = \frac{Z_{1-\alpha/2}^2 [P_1(1-P_1) + P_2(1-P_2)]}{e^2}$$

- Hypothesis tests for two population proportions

- For a one-sided test

$$\frac{\left[Z_{1-\alpha} \sqrt{2\bar{P}(1-\bar{P})} + Z_{1-\beta} \sqrt{P_1(1-P_1) + P_2(1-P_2)} \right]^2}{[P_1 - P_2]^2},$$

$$\text{where } \bar{P} = \frac{P_1 + P_2}{2}$$

- For a two-sided test

$$\frac{\left[Z_{1-\alpha/2} \sqrt{2\bar{P}(1-\bar{P})} + Z_{1-\beta} \sqrt{P_1(1-P_1) + P_2(1-P_2)} \right]^2}{[P_1 - P_2]^2},$$

- For a one-sided test for small proportion

$$n = \frac{[Z_{1-\alpha} + Z_{1-\beta}]^2}{[0.00061(\arcsin\sqrt{P_2} - \arcsin\sqrt{P_1})^2]}$$

- For a two-sided test for small proportion

$$n = \frac{[Z_{1-\alpha/2} + Z_{1-\beta}]^2}{[0.00061(\arcsin\sqrt{P_2} - \arcsin\sqrt{P_1})^2]}$$

...Summary

- Estimating an odds ratio with specified relative precision

$$n = \frac{Z_{1-\alpha/2}^2 \left[\frac{1}{P_1^*(1-P_1^*)} + \frac{1}{P_2^*(1-P_2^*)} \right]}{[\log_e(1-e)]^2}$$

- Hypothesis tests for an odds ratio

$$n = \frac{[Z_{1-\alpha/2} \sqrt{2P_2^*(1-P_2^*)} + Z_{1-\beta} \sqrt{P_1^*(1-P_1^*) + P_2^*(1-P_2^*)}]^2}{[P_1^* - P_2^*]^2},$$

- In this formula the term $2P_2^*(1 - P_2^*)$ is used instead of $2\bar{P}_2^*(1 - \bar{P}_2^*)$ because the study population is likely to be made up of many more controls than cases, and the exposure rate among the controls is often known with a high degree of precision; under the null hypothesis this is the exposure rate for the cases as well. If the investigator is in doubt about the exposure rate among the controls, however the formula should be modified and the term $2\bar{P}_2^*(1 - \bar{P}_2^*)$ used, where $\bar{P}^* = \frac{P_1^* + P_2^*}{2}$

...Summary

- Estimating a relative risk with specified relative precision

$$n = \frac{Z_{1-\alpha/2}^2 \left[\frac{1-P_1}{P_1} + \frac{1-P_2}{P_2} \right]}{[\log_e(1-e)]^2}$$

Since $RR = \frac{P_1}{P_2}$, $RR = \leq \frac{1}{P_2}$. For $RR = \leq 1$, use the column value corresponding to $\frac{1}{RR}$ and the row value corresponding to P_1 .

- Hypothesis tests for a relative risk

$$\frac{\left[Z_{1-\alpha/2} \sqrt{2\bar{P}(1-\bar{P})} + Z_{1-\beta} \sqrt{P_1(1-P_1) + P_2(1-P_2)} \right]^2}{[P_1 - P_2]^2},$$

where $\bar{P} = \frac{P_1 + P_2}{2}$

...Summary

- Estimating an incidence rate with specified relative precision

$$n = \left[\frac{Z_{1-\alpha/2}}{e} \right]^2$$

- Hypothesis tests for an incidence rate

- For a one-sided test

$$n = \frac{[Z_{1-\alpha} \lambda_0 + Z_{1-\beta} \lambda_a]^2}{[\lambda_0 - \lambda_a]^2}$$

- For a two-sided test

$$n = \frac{[Z_{1-\alpha/2} \lambda_0 + Z_{1-\beta} \lambda_a]^2}{[\lambda_0 - \lambda_a]^2}$$

- Hypothesis tests for two incidence rates in follow-up (cohort) studies (study duration not fixed)

- For a one-sided test

$$n_1 = \frac{[Z_{1-\alpha} \sqrt{(1+k)\bar{\lambda}^2} + Z_{1-\beta} \sqrt{(k\lambda_1^2 + \lambda_2^2)}]^2}{k[\lambda_1 - \lambda_2]^2}$$

- For a two-sided test

$$n_1 = \frac{[Z_{1-\alpha/2} \sqrt{(1+k)\bar{\lambda}^2} + Z_{1-\beta} \sqrt{(k\lambda_1^2 + \lambda_2^2)}]^2}{k[\lambda_1 - \lambda_2]^2}$$

where $\bar{\lambda} = \frac{\lambda_1 + \lambda_2}{2}$ and k is the ratio of the sample size for the second group of subjects n_2 to that for the first group n_1 .

Sampling distribution

- The sampling distribution of a statistic is the probability distribution of all possible values the statistic may assume, when computed from random samples of the same size, drawn from a specified population.
- Sampling distribution is a statistic that determines the probability of an event based on data from a small group within a large population.
- Its primary purpose is to establish representative results of small samples of a comparatively larger population.
- Since the population is too large to analyze, you can select a smaller group and repeatedly sample or analyze them.
- The gathered data, or statistic, is used to calculate the likely occurrence, or probability, of an event.

...Sampling distribution

Factors that influence sampling distribution

- We can measure the sampling distribution's variability either by standard deviation, also called standard error of the mean, or population variance, depending on the context and inferences you are trying to draw.
- There are three primary factors that influence the variability of a sampling distribution. They are:
 - The number observed in a population: The symbol for this variable is "N." It is the measure of observed activity in a given group of data.
 - The number observed in the sample: The symbol for this variable is "n." It is the measure of observed activity in a random sample of data that is part of the larger grouping.
 - The method of choosing the sample: How you chose the samples can account for variability in some cases.

Types of Sampling distribution

There are three standard types of sampling distributions in statistics:

- Sampling distribution of mean
 - The most common type of sampling distribution is of the mean.
 - It focuses on calculating the mean of every sample group chosen from the population and plotting the data points. The graph shows a normal distribution where the center is the mean of the sampling distribution, which represents the mean of the entire population.
- Sampling distribution of proportion
 - This sampling distribution focuses on proportions in a population. You can select samples and calculate their proportions.
 - The mean of the sample proportions from each group represent the proportion of the entire population
- T-distribution
 - A T-distribution is a sampling distribution that involves a small population or one where you don't know much about it.
 - It is used to estimate the mean of the population and other statistics such as confidence intervals.
 - The T-distribution uses a t-score to evaluate data that wouldn't be appropriate for a normal distribution.

...Sampling distribution

Relationships between Population Parameters and the sampling Distribution of the Sample Mean

- The expected value of the sample mean is equal to the population mean:

$$E(\bar{X}) = \mu_{\bar{X}} = \mu_x$$

- The variance of the sample mean is equal to the population variance divided by the sample size:

$$V(\bar{X}) = \sigma_{\bar{X}}^2 = \frac{\sigma_x^2}{n}$$

- The standard deviation of the sample mean, known as the standard error of the mean, is equal to the population standard deviation divided by the square root of the sample size:

$$SD(\bar{X}) = \sigma_{\bar{X}} = \frac{\sigma_x}{\sqrt{n}}$$

Sampling distribution of mean

Examples

- Example 1: Suppose we have a hypothetical population of size 3, consisting of three children: A is 3 years old, B is 6 years old and C is 9 years old. Construct sampling distribution of the sample mean of size 2 using sampling without replacement and with replacement.

Solution:

- The mean and variance of the population are 6 and 6, respectively.
- If sampling is without replacement we will have $3C2 = 3$ possible samples. $E(\bar{X}) = 6$ and $V(\bar{X}) = 3$
- If sampling is with replacement we will have $Nn = 3 \times 3 = 9$ possible samples. $E(\bar{X}) = 6$ and $V(\bar{X}) = 3$

Sampling distribution of mean

Example 2: Let $\bar{X}$ be the mean of a random sample of size 50 drawn from a population with mean 112 and standard deviation 40

- Find the mean and standard deviation of $\bar{X}$

Solution:

$$\mu_{\bar{X}} = \mu = 112$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{50}} = 5.65685$$

Sampling Distribution of the Sample Proportion, $\hat{p}$

- If repeated random samples of a given size n are taken from a population of values for a categorical variable, where the proportion in the category of interest is p , then the mean of all sample proportions $\hat{p}$ is the population proportion (p).
- The distribution of the sample proportion approximates a normal distribution under the following 2 conditions.
 - 1 $np \geq 15$
 - 2 $n(1 - p) \geq 15$

...Sampling Distribution of the Sample Proportion, $\hat{p}$

- If any set of the two conditions listed above are satisfied, the sampling distribution of the sample proportion is...
 - ① approximately normal
 - ② with mean, $\mu = P$
 - ③ standard deviation [standard error] $\sigma = \sqrt{\frac{p(1-p)}{n}}$
 - ④ If the sampling distribution of is approximately normal, we can convert a sample proportion to a z-score using the following formula:

$$Z_{cal} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

References

- ① Mukhopadhyay, Nitis. Probability and statistical inference/Nitis Mukhopadhyay. p. cm. (Statistics, textbooks and monographs; v. 162) Includes bibliographical references and index. ISBN 0-8247-0379-0 (alk. paper)
- ② An Introduction to Probability and Statistics by V.K. Rohatgi and A.K. Md. E. Saleh
- ③ Probability and Statistical Inference by Hogg, R. V., Tanis, E. A. and Zimmerman D. L.
- ④ Mathematical Statistics: A Textbook, S. Biswas and G.L.Sriwastav, Narosa
- ⑤ Cai J, Zeng D. Sample size/power calculation for casecohort studies. Biometrics 2004;60:101524.
- ⑥ S. K. Lwanga and S. Lemeshow. Sample size determination in health studies : A practical manual. World Health Organization, Geneva, 1991. [https://apps.who.int/iris/handle/10665/40062;9241544058\(p1 – p22\).pdf](https://apps.who.int/iris/handle/10665/40062;9241544058(p1-p22).pdf); [9241544058\(p23 – p80\).pdf](https://apps.who.int/iris/handle/10665/40062;9241544058(p23-p80).pdf)

Thank you !!!